

Cisco CCNA 200-301 - The Complete Guide To Getting Certified

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Chapter 1

Host to host communications

1.0.1 A very basic introduction to networking

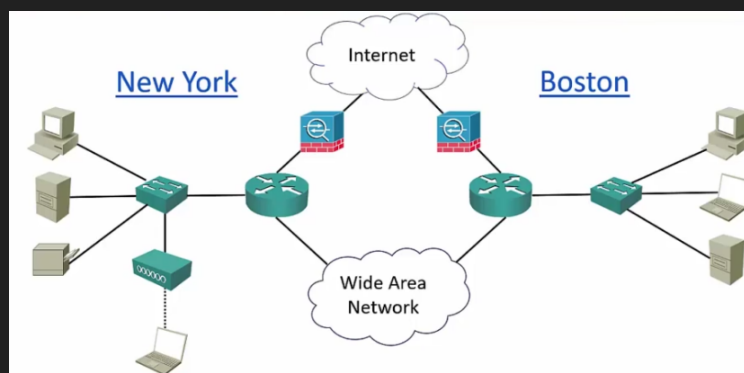
Let's say we have an office in NY. In that office there are end-hosts, such as PCs, server, printers, etc. All of the end hosts need to communicate with each other, for this we introduce a switch, this switch is going to be connected to each of the end-hosts using ethernet cables. A switch is a device with a lot of ports - which are basically slots to connect ethernet cables to - this allows the device to be connected to each other through the switch. The switch is what allows connectivity on my *local area network*. Let's say we add another end-host, a laptop, which is connected wirelessly to the same switch, the laptop will be connected via WIFI to a *wireless access point* and the wireless access point is connected to the switch via an ethernet cable. Now that all the end-hosts are able to communicate, we have built a *local area network*, which means a network that connects devices within the same local area, such as an office or a university campus.

Now that we have built a local area network, we also want these end-hosts to be able to communicate with other devices out on the internet, not just in my local area network. For this we introduce another component, the *router*, which is a device capable of making advanced routing decisions to route traffic between different areas of your network.

Next, now that we have connected the end-hosts to the local are network, and the local area network to the internet via a router. The internet is a very dangerous place, hackers can attack us and other things that we don't want can happen. For security reasons we install a *firewall*, which secures the different parts of our network from each other.

In a larger company we would not just have end-hosts connected to switches connected to routers connected to firewalls connected to the internet. We would have similar local area networks in many other places, for example, we can say that we have a very similar local area network in Boston. The same devices and the same connections. Lets say we want the NY and Boston office to communicate directly without using the internet. We can use a *dedicated connection* between my two routers on both sites for communications to be direct from the NY router to the Boston router, this is called a *wide area network*.

All this is expressed in the next, *network topology diagram*.



Some other characteristics of networks are:

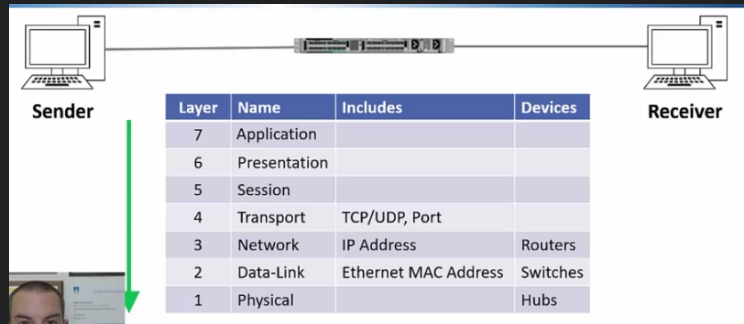
- Topology: how the devices are connected to the each other.
- Speed: the faster, the more cost.
- Cost: the monetary cost, effected by the type of devices and technologies used.
- Security: all the security features inside and outside the devices.
- Availability: making sure that our network always stays up. Mission critical components are typically duplicated, that way we can have an alternative if one were to fail.
- Scalability: the topology design lends itself to grow as needed.
- Reliability: similar to availability, making sure the network is going to continue to work.

1.1 The OSI Reference model

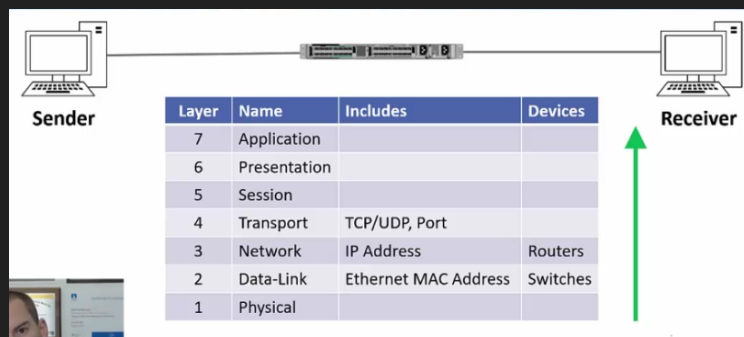
- The OSI model stands for Open Systems Interconnect model.
- It is standardized by the Organization for Standardization (ISO).
- It is a general purpose network that standardizes how computers communicate with one another over a network.
- The OSI model is conceptual it is not a physical thing, device, or protocol. It is a concept for humans only, to be able to understand.
- The 7 layer approach to data transmission divides the operations into specific related groups of actions at each layer. A layer serves the layer above it and is served by the layer below it.

1.1.1 Example

An example is how you send an email. The sender (the person sending the email) and the receiver (the person receiving the email). In the layer 7 - the application layer - which is the layer containing the “from:” or the “to:” field, this is collected at an application layer. Next the data will be encapsulated by the layer 6 - presentation layer - which deals with the format the data is being handled with. Then the layer 6 information is encapsulated into the layer 5 - session layer - information. That then gets encapsulated to the 4th layer - the transport layer - which is the layer that specifies what communication will be used - TCP, UDP, Port number - to prepare it for the next layer. That gets encapsulated into layer 3 - the network layer - which includes the source and destination IP address, a device that operates at layer 3 is the routers. The next the packet is almost being finished, and passed on to the layer 2 - the data link layer - which specifies the mac address - in case we are sending this over an ethernet network -, devices that operate at this layer are switches. Finally the encapsulated and consolidated packet is ready to be sent, this is done in layer 1 - which is the Physical layer - which literally means sending the packet through the wire, a network device that used to be used in this layer are hubs - which are not used anymore -. The email has been sent. The packets travel through the wires and devices to the receiver and the receiver receives packets in the form of electrical signals.



The receiver will process the information being received from layer 1 to layer 7, which means that the receiver will grab the data from the first layer and decapsulate it until it has layer 7 data. It comes in at the physical layer, then it will pass to the data link layer which is 2nd layer, it will perform checks to ensure that the destination mac address is really for the receiver, if it is not, the receiver will just discard the packet. Next it will look at the layer 3 - which is the network layer - it will check the destination ip address and check if the packet is for it, otherwise it will discard the packet. Then we carry on with the next layer, layer 4 - the transport layer - it will check the port number in which it was received and some other validations. It will continue through the session layer - layer 5 -, the presentation layer - layer 6 -, and the application layer - layer 7 -.



Layer 6 and 7 - application and presentation - are considered the upper level, typically used by developers not network engineers. Usually network engineers start to become interested from layer 4 downward.

1.1.2 Benefits of the OSI model

- Engineers don't need to design technologies to work end to end from top to bottom of the model. Instead they can just focus on their layer of expertise, and make sure they comply with the standards for the layers above and below.
- This leads to open standards and multi-vendor interoperability.
- For example: if you're an application developer, you can just focus on the top three layers, the lower layers are the domain of network engineers. You don't need to understand them, you just make sure you are complying with the standards.
- Troubleshooting is easier because you can analyse a problem in a logical fashion layer by layer.
- It is difficult to convey how important the OSI model is. The OSI model is used to describe problems, solutions, technologies, etcetera. Such as if the cable was disconnected you would say 'it was a layer 1 problem', if you find that the problem was that the user specified a wrong destination - you say - 'it was a layer 7 problem, etcetera.

1.1.3 OSI Acronyms

- The classic: Please Do Not Throw Sausage Pizza Away.
- Network Relevant: Please Don't Need Those Stupid Packets Anyway.
- Us relevant: Please Do Not Teach Students Pointless Acronyms.
- Useful: Please Do Not Take Sales People's Advice.
- Favorite: Please Do Not Touch Superman's Private Area.

1.2 TCP/IP Stack

- TCP/IP was developed during the 1960s by the US department of defense's Advanced Research Projects Agency (ARPA).
- It is a protocol stack which consists of multiple protocols including TCP (Transmission Control Protocol) and IP (Internet Protocol). It is not a single protocol, it is a protocol stack.
- An analogy to understand what a protocol is is when diplomats meet each other, there is a certain way they are expected to behave and to communicate with each other, in computer networking protocols basically mean the same thing. You have two hosts and they are going to communicate with each other, there has to be a protocol that describes that communication is going to be and how is it going to behave.
- It is the main protocol stack used in computer operations today.
 - There used to be many other protocols such as Ipxspx and apple talk. But today TCP/IP is pretty much standard, other protocols are pretty much dead.
- Whereas the OSI reference model is conceptual, the TCP/IP stack is actually used to transfer data in production networks.
- TCP/IP is also layered but does not use all the OSI layers, though the layers are equivalent in operation and function.
 - It does actually use all of the layers, but in the documentations only 4 are layed out.

1.2.1 Comparing the OSI model with the TCP/IP

OSI Model	TCP/IP Stack	Cisco TCP/IP Stack Layer Definition
Application	Application	'Represents data users, encodes and controls the dialog.'
Presentation		
Session		
Transport	Transport	'Supports communication between end devices across a diverse network'.
Network	Internet	'Provides logical addressing and determines the best path through the network'.
Data Link	Network Access	'Controls the hardware devices and media that make up the network'.
Physical		

Notice that the TCP/IP does cover all the layers of the OSI model, however it sometimes uses the layers together. The application layer - in the TCP/IP - maps to the OSI model's layer 5,6, and 7 - session, presentation, and application -. Below that we have the transport layer - in the TCP/IP stack - which notice

maps directly to the transport layer in the OSI model. The internet layer in the TCP/IP is equivalent to the network layer in the OSI model. Last we have the network access in the TCP/IP stack that maps to the OSI layer 1 and 2 - physical and data-link-.

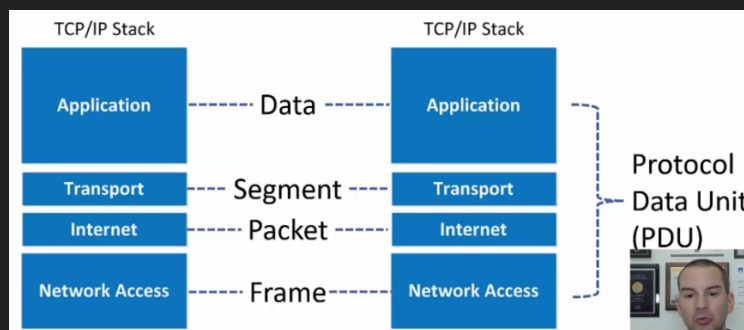
- While both the OSI model and the TCP/IP stack are very important in networking. We are going to typically be referencing more the OSI model than the TCP/IP stack.

1.2.2 Hosts communication terminology

When two hosts communicate with each other, they talk using something called protocol data unit (PDU), which is the entire communication that happens on the OSI model (sender from layer 7 to layer 1 and receiver from layer 1 back to layer 7).

The data exchanged varies in terminology according to which layer of the TCP/IP stack is being talked about.

- The application layer communicates using data to the application layer in the other host.
- The transport layer communicates using segments to the transport layer in the other host.
- The internet layer communicates using packets to the internet layer in the other host.
- The network access layer communicates using frames to the network access layer of the other host.



In the networking world the OSI model is more commonly referenced than the TCP/IP stack in terms of layers - which means that it is more common to refer to the layers of the OSI model than the layers of the TCP/IP -. However when we are talking about host communications terminology, it is more common to speak of the TCP/IP communications terminology - which means that it is more common to use 'data' if you are referring to the layer 5,6, and 7 of the OSI model, layer 3 call it a packet, layer 4 call it a segment-. Another very important thing is that networking people usually refer to the PDU as the name packet, the correct name is a PDU not a packet since the packet is used for layer 2 (internet layer) communication, but people typically use the term packet and PDU interchangeably, while technically wrong, but in day-to-day communications when referring to host communications it is not uncommon to call it packet meaning the entire stack.

1.3 The upper OSI layers

- The networking engineers do not typically work directly with the upper 3 layers of the OSI model, but we still need to know what they do.
- They are more relevant to application developers.
- In this lecture I will primarily be giving you the Cisco definitions of the layers.
- Information included in the upper layers would include the message body and subject line in an email message for example.

1.3.1 Layer 7 - the application layer

- The application layer provides network services to the applications of the end user.
- It differs from the other layers in that it does not provide services to any other OSI layer.
- The application layer establishes the availability of intended communications partners.
 - Intended communication partners are the hosts that the current hosts is trying to communicate with.
- It then synchronizes and establishes agreement on procedures for error recovery and control of data integrity.
 - Data integrity is verifying that data has not been corrupted or damaged in transit.

1.3.2 Layer 6 - the presentation layer

- The presentation layer ensures that the information that is sent at the application layer of one system is readable by the application layer of another system.
- The presentation layer can translate among multiple data formats using a common format (eg computers with different encoding schemes)

1.3.3 Layer 5 - session layer

- The session layer establishes, manages, and terminates sessions between two communicating hosts.
- The session layer also synchronizes dialog between the presentation layers of the two hosts and manages their data exchange.
- For example, web servers have many users, so there are many communication processes open (session) at any given time to track. The session layer keeps track of all of the opened sessions.
- It also offers efficient data transfers, CoS (Class of Service), and exception reporting of upper layer problems.

1.4 The lower OSI layers

- Whereas network engineers are not particularly interested in the upper OSI layers, we are very concerned with the lower 4 layers of the OSI model. These are incredibly important and are almost like the daily routine to network engineer.
- Each of these layers have their own dedicated section later and you will learn much more detailed information about them throughout the course.

1.4.1 Layer 4 - transport layer

- The main characteristics of the transport layer are whether TCP or UDP transport is used, and the port number.
- We haven't really seen what is TCP and UDP, for now you can think of it as modes of transportation where each one has different advantages. The TCP is very reliable, therefore if priority is reliability we use TCP. If speed is more important than reliability - like for voice or video traffic - then we use UDP instead.
- We haven't seen what ports are, but for now you just need to know that different processes occur in these ports, for example the 80th port is used for http web traffic and port 25 is used for email.

- Definition:
 - The transport layer defines services to segment, transfer, and reassemble the data for individual communications between the end devices.
 - It breaks down large files into smaller segments that are less likely to incur transmission problems.

1.4.2 Layer 3 - network layer

- The most important information at the network layer is the source and destination IP address. Other information is also carried, but the main is the source and destination of IPs.
- Routers operate at layer 3.
- Definition:
 - The network layer provides connectivity and path selection between two host systems that may be located on geographically separated networks.
 - The network layer is the layer that manages the connectivity of hosts by providing logical addressing (IP addressing is our logical addressing).

1.4.3 Layer 2 - the data-link layer

- The most important information at the data-link layer is the source and destination layer 2 address. Again - just like in layer 3 and 4 - there is also other information but the main one is the source and destination layer address.
- For example the source and destination MAC address if ethernet is the layer 2 technology. Different layer 2 technologies use different formats for their addressing, for example, old legacy frame relay uses DLCI or DCE numbers for the addressing. Today the typical address used at this layer is the MAC address.
- Switches operate at layer 2.
- Definition:
 - The data link layer defines how data is formatted for transmission and how access to physical media is controlled.
 - It also typically includes error detection and correction to ensure a reliable delivery of data.

1.4.4 Layer 1 - the physical layer

- The physical layer concerns literally the physical components of the network, for example the cables being used.
- Definition:
 - The physical link enables bit transmission between end devices.
 - It defines specifications needed for activating, maintaining, and deactivating the physical link between end devices.
 - For example, voltage levels, physical data rates, maximum transmission distances, physical connectors, etcetera.

Chapter 2

The cisco IOS Operating System

2.1 Introduction

By now we all want to configure the cisco devices such as routers and switches. Cisco devices do include GUIs - which means graphical user interface - but in the real world, everybody just uses the command line interface (CLI) because it is much quicker than using the GUIs. The OS used in enterprise grade cisco devices is IOS, in this section we'll learn how to connect to the devices and the use of the command line. A disclaimer is that the IOS operating system is not the only one, there are others.

2.2 Cisco operating systems

Cisco history is not going to be examined on the CCNA exam. Neil includes this just because of context. Cisco IOS and the other OS's work basically the same way in terms of management and administrative point of view, the operating systems don't differ that much in terms of how you use it, maybe on how they are actually implemented but not really on how you use it. By learning cisco IOS you are really learning all of the others at once.

2.2.1 A short history of cisco operating systems

- Most people think of cisco as primarily a routing and switching company, but they actually started out with just routers in 1984.
- IOS was the original operating system that they used on routers - the same OS they use today (with some optimizations).
- IOS is the operating system that has been used on cisco routers since their inception.
- Cisco Catalyst switches evolved from the acquisition of Crescendo company in 1993 - cisco bought the company Crescendo- and started manufacturing switches of which the first one was Catalyst.
- The original cisco switch operating system was CatOS which has not been deprecated.
- Cisco firewalls evolved from the acquisition of the company Network Translation's PIX firewall that uses the Finesse operating system in 1995.
- Cisco switches and firewalls (now the ASA firewall) were ported over the IOS operating system over the following years.
- Cisco standardized on IOS for all the network infrastructure devices.

There are some other operating systems in some newer router and switch platform. IOS runs on the majority of routers and switches. Some of these other operating systems are:

- The Cisco Nexus and MDS data center switch product lines run on NX-OS.
- The IOS-XR operating system runs on the service provider NCS, CRS, ASR9000, and XR12000 series routers (these are the higher end routers).
- IOS-XE runs on the ASR1000 series service provider routers (1K service providers).
- The command line interfaces for the other operating systems are nearly identical to IOS.

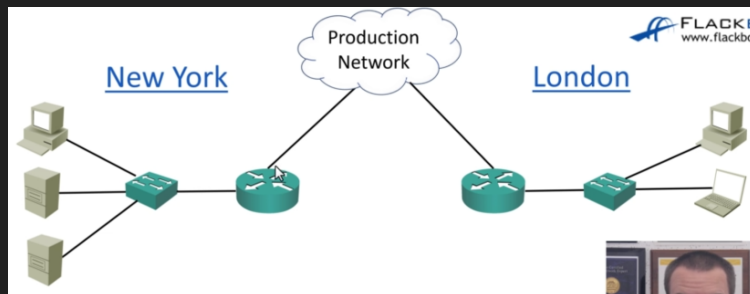
At first all of these names might seem intimidating but fear not, all of the OS's run nearly identical commands, if you can use one, you can practically use them all. There are different OS's because they are implemented with different functionality, for example, IOS has a monolithic kernel - which means that if one process crashes it crashes the entire router - they are all different under the hood. Other OS's have different kernels to permit that if one crashes the other processes in the router are not affected, only that process is crashed. This does not mean that IOS is bad or unreliable, it simply means that it works differently given different situations.

2.3 Connecting to a cisco device over the network

- The lab exercises in this course use Cisco Packet Tracer simulation software on your PC.
- This lecture shows how to connect to a real router or switch over the network with Putty - which is a software to connect to routers - you are not going to use Putty in the labs because everything is done in the Packet Tracer simulation software.
- You do not need to install or use Putty to do the course lab exercises.

2.3.1 Connecting over the network

In your day-to-day as a network engineer you are typically not going to be connecting stuff, instead you are going to be in your desk 'working remotely', you are going to be connecting to that device over the network. So the following example: Let's say that you are in a London office and want to establish a connection in the NY office. What you are going to do is use software on your PC, connect to the router over the network in NY, get to the command line of device you want to work on, and get on with your job.

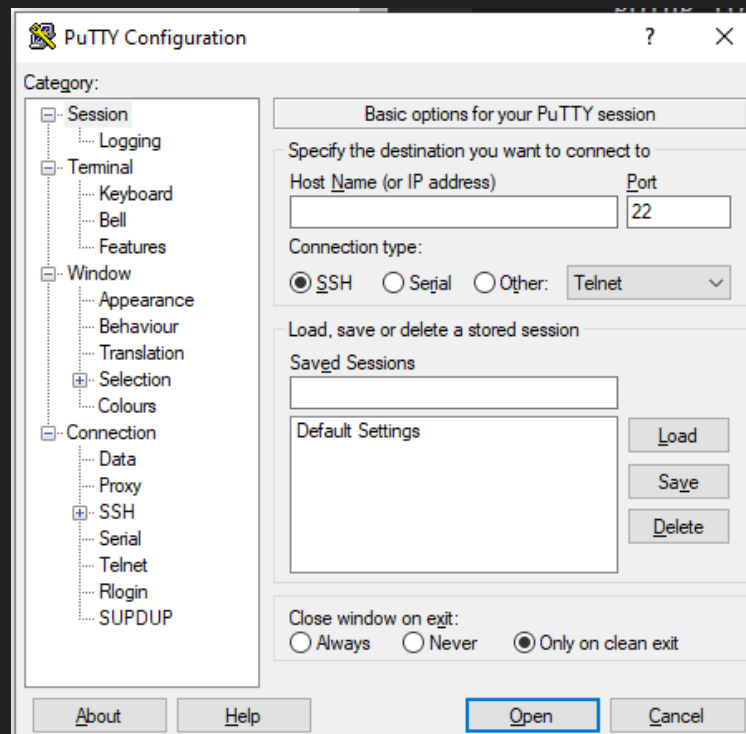


- To get to the CLI for day-to-day management of a Cisco device you will use Secure Shell (SSH) to connect to its management IP addresses over the network.
- There is also another protocol called Telnet which is also supported but not recommended because it is insecure. Commands in Telnet are un-encrypted, so anybody listening in to your traffic can get ahold of data, including username and password. Telnet is never used in a real setting for the potential vulnerabilities. In reality SSH is always used, Telnet is almost always used only on a lab environment to learn.
- In enterprise networks, secure login will typically be enforced through integration with a centralized AAA (Authentication, Authorization, and Accounting) server.

- Basically the AAA server is a solution to many people logging in. In a network, you can probably have many devices with username and passwords that must be verified to be granted access to network actions. We don't want each device to have the username and passwords of all the others so we centralize that into one server that stores all the user name and passwords that has access to the central database of usernames and passwords.

2.3.2 Installing Putty

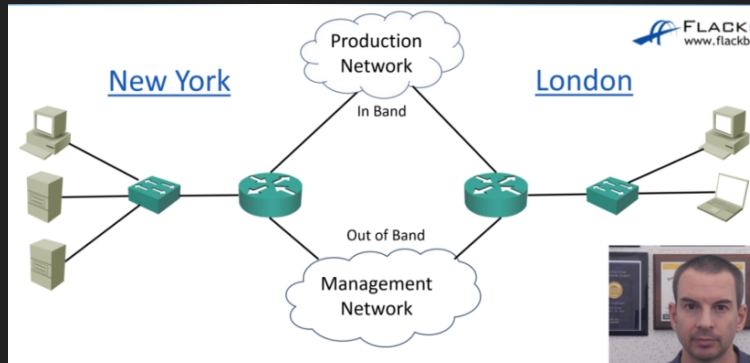
Installing Putty is basically just searching on Google and accepting all the defaults once the executable has been downloaded. When you launch the window you will be faced with a window like this:



The host name refers to the host that I am intending to connect to, it takes as an input an IP address. Then you log in through the command line, and type your password. This is how you connect to a cisco device, very simple using Putty.

2.3.3 Out of band management

To continue the example in which we were connecting in London to the NY office, we are essentially using the production network. But most large companies use a separate management network, this is done as a backup path in case the production network is down, it is also more secure to have a separate dedicated path. The production network is the in band network, which means you are using the same cables that your normal users (your staff) are using to connect to other devices. If you connect to the NY office through the management network you are using and out of band network, which means you are not using the same network as the normal staff. Trivial difference but important distinction.



2.4 Making the initial connection to a cisco device

- Cisco devices do not usually have a default IP address, so we need to set one up before we can connect to it over the network, but how do we connect to the device in the first place if we don't have an IP?
- We need a way to connect to the device to do the initial configuration including adding IP addresses. This is where the console connection comes in. With a console connection we can connect to the device at a lower level below IP to get to the initial command line to do the initial configuration including adding an IP and finally be able to connect to it over the network.

For this initial configuration, we need a Console Cable (DB9 to RJ45) in order to connect to the router via a wire and send the configurations. Simply connect the cable to the console port on the router or switch and use the cable - which usually comes with your device.



Figure 2.1: Console Cable DB9 to RJ45

The console cable has two ends, one is the DB9 which is the thicker end, it is called that because it has 9 pins need to be connected to that end.



Figure 2.2: DB9 end

The other end is very similar to an ethernet cable. However, **THIS IS NOT AN ETHERNET CABLE.** It is a RJ45, it is not using ethernet and it doesn't require IP addresses, this only grants you low level access to the command line.



Figure 2.3: RJ45

When you connect to make the initial configurations to the router using the console cable, the RJ45 - the one that looks like an ethernet connection but is not - end of the cable is the one that is going to be connected to the switch. The other end - the serial connector - plugs in to the computer. This is a problem for computers that don't include the DB9 connector, for this you can buy an adaptor from DB9 to USB.



2.4.1 Newer console cables

With newer cisco devices the cable included comes already with a USB end for the computer since most computers don't come with DB9 cable support. So the newer console cables come with some modifications, not only the USB end for the computer but they actually removed the RJ45 ethernet looking but not ethernet end to be instead a microusb end.

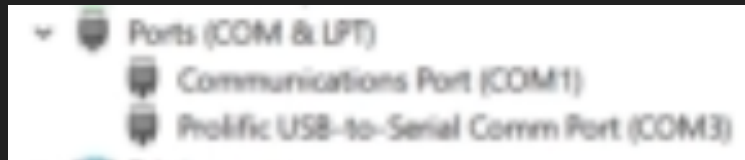


Figure 2.4: USB to microusb

2.4.2 Connecting via console cable to a real router

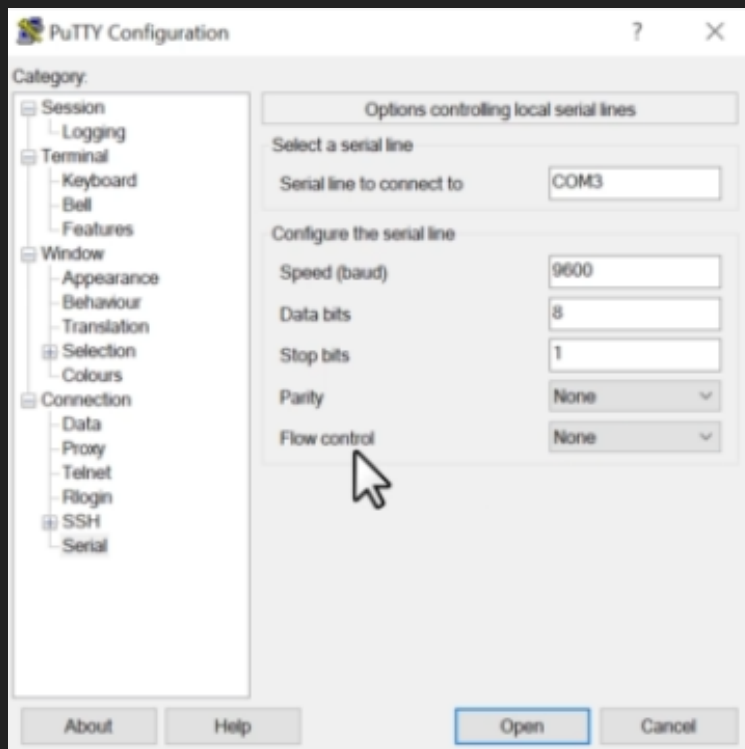
First identify the console port on the router and connect the appropriate end (microusb or RJ45) to the router, the other end to your computer (USB or DB9). Launch putty and click that the connection type is going to be 'Serial'. Now you are going to have to specify your COM port correctly, if you don't know which one it is follow the following steps (windows):

- Make sure that you have installed the driver software for the console cable in order for the COM port to be enabled.
- Open device manager and look in 'Ports (COM & LPT)'.



- In this case it is COM3, change in putty the COM number port specified in the device manager.

Then you need to select the speed, specified in the 'Serial' tab, and set flow control to None.



Return to the 'Session' tab and click on 'Open'. Now turn on your router, you will start to get output on the screen. The next thing is that the router will boot up, this will take some time so give it some time.

2.4.3 Console connection troubleshooting

- When we buy a new device we saw how we can make the initial configurations including adding the IP addresses in order to connect to it over the network.
- As well as for initial configuration, the console port - and cable - can be used if the device's IP addresses become unresponsive.
- It can also be used to troubleshoot the bootup process. You can view the device booting up from a console connection but this is not possible with SSH because the system must have booted already before the IP addresses will be live.

- Another case is when a device just appears to be completely unresponsive but you can see that the device is on, that indicates that the device is probably failing to boot up. Again trying to connect to it over the network will not work, you need to take out your console cable and troubleshoot it. It will not work by connecting to it because the IP is only live until it has completed the bootup process, therefore you cannot connect it through the network, but you can connect to it over the cable and power the device on and you can watch if any problems ensue.

2.5 Navigating the cisco IOS Operating System Part 1

2.5.1 Booting up, and prompt initial configurations

Once you buy a router and connect the console cable to it, output will be displayed on the screen of the PC you connecting the other end of the console cable to. This means that the router is booting up. Once it has finished booting up a prompt will appear and ask you if “Would you like to enter the initial configuration dialogue? [yes/no]: ”. If I say yes, it will take me to a command line driven wizard which would help me do the initial setup. However this option is not usually used, so we will say no. Once you say no, another prompt will appear “Would you like to terminate autotinstall? [yes]: ” type yes. Once you have said yes to the previous prompt the command line will take you to the user exec prompt, however this command line environment is very limited in the number of commands we can process. If you want to see which commands are supported in the exec prompt type “?”. This will print some of the commands, the 'More' word at the bottom signals that there are more commands, to see the rest of the commands we can hit the enter key and it will display the new commands line by line, by using the space bar you can display the commands page by page.

```
Router>?
Exec commands:
access-enable      Create a temporary Access-list entry
access-profile     Apply user-profile to interface
clear              Reset functions
connect            Open a terminal connection
credential          load the credential info from file system
crypto             Encryption related commands.
disable            Turn off privileged commands
disconnect          Disconnect an existing network connection
do-exec            Mode-independent "do-exec" prefix support
emm               Run a configured Menu System
enable             Turn on privileged commands
ethernet           Ethernet parameters
exit               Exit from the EXEC
help               Description of the interactive help system
ip                 IP SLA Exec Command
ips                Intrusion Prevention System
lat                Open a lat connection
liq                LISP Internet Groper
lock               Lock the terminal
login              Log in as a particular user
logout             Exit from the EXEC
-- More --
```

Now, as you continue to type commands and output is printed to the screen, some previously typed lines are going to be erased, to save them you hit on Putty reconfiguration, go to the window tab, in the attribute called “Lines of scrollbar: ” set it to a large number such as 2,000.

2.5.2 Switching from user exec prompt to privileged exec mode

As we said the exec prompt is very limited and we can't do much in this prompt, so we are going to switch to another prompt called the privileged exec mode, to go to a privileged exec mode simply type “enable” or it's abbreviation “en”. Then the command prompt changes, now instead of each line being something like “Router>” you have something like “Router#”. To be switch from privileged exec mode to normal exec user mode simply type “Disable” or it's abbreviation “disa”.

```
Router>enable
Router#disable
Router>en
Router#di
```

2.5.3 Ambiguous abbreviations

Abbreviations for command need to map uniquely to one command. For example we can abbreviate the “enable” command to just “en” because it is the only command that starts with “en” therefore that abbreviation is not ambiguous. With the “disable” command we cannot abbreviate as “di” because it is not the only command that starts with “di” other commands such as “dir” and “disconnect” also starts with “di”, therefore to make it unique we not only need to add an “s” to “di” but also an “a” to make it “disa”. To see if an abbreviation is ambiguous you can type the abbreviation followed by a question mark. Such as: `Router# di?`. It is important that when you are checking this the question mark is included after the proposed abbreviation without a space because a space and then question mark is something else.

2.5.4 The show command, tab completion and errors

The “show” command - or it’s abbreviation “sh” are used to be able to see information inside the kernel. When you type in abbreviation followed by a space followed by a question mark, the kernel will return all the possible keywords that are available for that particular command, this is known as context sensitive help - which means which combinations of commands we can use together. If we type “sh ?” it will print all keywords that can be used with the “sh” command before it. If we say something like “sh aaa ?” which are both commands, are asking the kernel to tell us which other commands can go after a command that is of the structure “sh aaa”. Also abbreviations can be expanded to the full equivalent by using tab, so for example entering a command such as “acct-stop-cache” you can instead type in “acc” and press tab and the kernel will insert the full equivalent of “acct-stop-cache”, if the word is ambiguous you will probably need to add more characters to make it unique. If you make a typo and hit tab the kernel will print an error saying that the command expansion is invalid, and a marker ^ will indicate on which letter the command failed. If you do an incomplete command then an error will be printed saying that the command was incomplete.

2.5.5 Breaking out of commands with ctrl+c

If we type a command and we want to break out of it, by simply typing ctrl+c we can break out of that command and any process being executed by that command is now canceled.

2.5.6 More on the enable prompt and debug

The enable prompt is the prompt where we will most likely be working on. There are commands that allow you to debug information or processes. You can see all available debug commands by typing “debug ?” with a space - because we need a context sensitive help -. This will allow us to see information and see if they are running as intended. You cannot make changes to such information in the privilege exec prompt, to make changes you need to enter to another prompt called the global configurations prompt.

2.5.7 The global configurations prompt

To actually correct and change how processes and configurations are set we need to go to the global configuration which you can go there by typing “configure terminal” or its abbreviation “conf t”. To see if you have entered into the other prompt correctly, the lines will now start with something like “Router(config)#”

2.6 Navigating the cisco IOS operating system Part 2

2.6.1 Command history

Lets enter a couple of random commands - you don’t need to understand the commands - just to create some commands in order to be able to visualize the history.

```
1 -> Router(config)#ip host Server1 1.1.1.1
2 -> Router(config)#ip host Server2 2.2.2.2
3 -> Router(config)#Router1
      ^
% Invalid input detected at the '^' marker.
```

See the commands we typed in, the first one was ok, the second was ok, but the third one has an error. We need to fix the typo in command 3. We can surely type the command again and problem solved, but imagine it is a gigantic command and just one typo got an error and typing it again is going to be a waste of time. By simply hitting the up-arrow-key the last command will be re-written in the current line, there you can correct the typo and enter. If the typo is at the beggining of the command, to get to the beggining in one step you can type ctr+A which will put the mouse cursor in the begging of the line.

The command history saves all the commands you’ve typed in, it does not matter if they worked or not, they are saved in the command history. Therefore, by pressing the up-arrow-key you can put previous commands in the current line, and by pressing the down arrow key you can navigate downward through the history. The command history only shows the commands in the global configuration prompt, not on the enable prompt or in the user exec mode.

Remember the commands are valid depending on the command prompt level, you cannot do a command that is intended to be only runned in a certain prompt - such as the user exec prompt, etcetera - and then expect to use the same command in global configuration prompt or some other prompt. The command history comes in handy not just for typos but also if you make a mistake and type in a command at the wrong prompt level, therefore you can get to the right level and then by hitting the up-key you can run the same command and you don’t waste time. For example:

```
Router(config)#hostname Router1
Router1(config)#show ip interface brief
      ^
% Invalid input detected at the '^' marker.
```

We are at the wrong prompt layer, show commands need to have the proper privileges, we can give the command proper privileges by saying “do” at the beggining - remember you are not supposed to know what these commands do right now, it is just for show -. Press the up key and this will appear again:

```
Router1(config)#show ip interface brief
```

Put ctrl+a to go the the beggining of the line and type in “do”.

```
Router1(config)#do show ip interface brief
```

Now it should work.

2.6.2 Prompts so far

So far we have seen that 3 prompts exists, and sort of seen how they are used and what they do.

1. User exec mode prompt
2. The privileged exec mode - also known as the enable prompt -
3. The global configuration: used to set configurations for things that affect the router as a whole.

There are other levels, such as the interface configuration level which you can get to by typing:

```
Router1(config)#interfase fast fastEthernet 0/0
Router1(config if)#
```

2.6.3 Exiting pompts or 'dropping levels'

If you want to drop down a level you can say "exit" command, this will take you to the previous level. If you want to drop all the way back to the enable prompt you can simply type in the "end" command or by hitting the ctrl+c command.

2.6.4 Most common commands

- **show ip interface brief** : shows all the interfaces on the router, and the ip addresses as well as the protocol.
- **show running config** : shows the entire configuration on the router. Remember by hitting the space bar you can scroll through pages since this command does print out a lot of text.
- **sh run int fast0/0** : shows the configurations of a specified interface (int this case interface 0/0).
- **sh run | begin hostname**
: this command use a 'pipe' (which we haven't seen), you can see if a command use a 'pipe' if the pipe symbol - which is '|' - is present in the command. In general, you can see the context of this command by simply putting a space and then a question mark after the pipe symbol, in this case the 'begin hostname' is simply to display the configurations running on that specific host name.

2.6.5 The IOS command line is not case sensitive except for one thing

Take into account that IOS is not case sensitive except for 'pipes' which use regular expression which is case sensitive. An example is:

```
Router1#sh run | begin Hostname
Router1#
```

After hitting enter to run the command the command printed no output, this is because 'Hostname' with a capital letter 'H' does not exist in the router so there is no output to show running configurations from. Pipes use REGULAR EXPRESSIONS which you don't really need to know anything about yet except that they ARE CASE SENSITIVE. This is the only exception to the IOS command line. Pipes are the only exception to the IOS command line.

2.7 Cisco IOS Configuration management

In this section we will learn about IOS configuration management and running/startup configuration. It is important to realize that in IOS everytime you make a change the change takes effect immediately, the change goes into the running configuration which takes effect immediately, this is the running configurations which take in commands and go into effect immediately. Aside from the running configurations we also have

the startup configurations - which are different from the running configurations - the startup configurations changes take effect when the router is restarted or rebooted, commands in the running config do not get saved permanently until you explicitly save them to the startup configuration, this is done so that you only make the changes take effect if you are sure they are correct. For example if you have a router and send configurations that are catastrophic but you have not saved it yet, you can simply restart the router and the changes that were catastrophic would simply never had taken place, this is a last resort.

2.7.1 Example of saving to startup configurations

Make sure you are in the user privilege exec prompt to do this, type “copy run start” and the kernel will ask you to enter a destination filename, it is common practice to stick with the default destination filename, and now if you hit “sh start” which means to show the startup configurations the changes were saved permanently.

2.7.2 Backing up configurations

*Sidenote: cisco does have an automated way of backing up configurations from a centralized server, but you can also backup manually as well. Simply type:

```
Router1#copy run ?
Router1#copy run flash:my-configuration
Destination filename [my-config]?
935 bytes copied in 1.236 secs (756 bytes/sec)
```

In this case we backed it up to a special location inside the router that is a flash drive location. If something were to happen we can retrieve our configurations from the configuration file backed up in the flash. To see if the configurations are indeed in the flash we can type the following command: “sh flash” (the name of the configuration file should be there). If we were to want to restore a back up we simple need to copy the back up to the start up config, but when you do a copy it is going to actually merge the commands rather than replace so typically you are going to want to replace the entire startup configuration, to do this you simply need to erase the startup configuration using the “erase start” - in newer routers and “wr erase” in older routers - once the startup config has been erased you can simply use the “copy flash:my-config start” to copy the backup to replace the startup configuration file.

2.7.3 Backing up outside the router

If the whole router blows up we would loose everything including the flash configurations, therefore it is a good idea to store those configs in a place where we can retrieve them even if the router blows up or anything of the sort. For this we can use a tftp server to backup the configurations. For this type the following commands:

```
Router1#copy run tftp
Address or name of remote host []? 10.10.10.10
Destination filename [router1-config]? router1-config-111111
```

The tftp server commands require that you have a tftp server to back up configurations, enter the IP address of the tftp server you have and then type in the name of the file you want, typically it is a good idea to save it as the routename followed by config followed by the date the backup is being performed.

2.7.4 How to check the contents of my backup configurations files

Use the the following command (replacing the configuration filename placeholder with your actual filename, in this case it would be “my-config”):

```
Router1#more flash:my-config
```

2.7.5 Configuration storage locations

- For the device to bootup it needs to load the IOS operating system image that is stored in flash.
 - The IOS operating system image is stored in Flash.
- Once the operating system is up and running you startup configurations will be loaded and processed.
 - The startup configuration is stored in NVRAM (non-volatile memory).
- The running configuration is stored in normal memory, which is RAM. It is loaded into RAM from the Startup Config when the device boots up.